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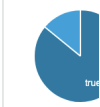
<https://github.com/antelcha/awsprojeler>

AWS ile ML Modeli Eđitme

Proje 3: Akıllı Veri Analitiđi ve Makine Öğrenmesi Uygulaması

Bu projede AWS SageMaker modeli kullanarak bulut üzerinde model eğittim.

Kullandığım veri seti house price prediction dataset.

About this file Suggest Edits					
The dataset has 13 features and 545 samples.					
# price Price of the Houses	# area Area of a House	# bedrooms Number of House Bedrooms	# bathrooms Number if Bathrooms	# stories Number of House Stories	✓ mainroad Weather c Main Road
 1.75m 13.3m	 1650 16.2k	 1 6	 1 4	 1 4	 true
13300000	7420	4	2	3	yes
12250000	8960	4	4	4	yes
12250000	9960	3	2	2	yes
12215000	7500	4	2	2	yes
11410000	7420	4	1	2	yes
10850000	7500	3	3	1	yes
10150000	8580	4	3	4	yes
10150000	16200	5	3	2	yes
9870000	8100	4	1	2	yes
9800000	5750	3	2	4	yes

Yapacağım proje lokalde önce modelin parametrelerini ve yapılandırmasını yapacak ama asıl gpu gerektiren kısmı SageMaker üzerinde yapacak ve orada eğitilen modeli her yerde kullanabileceğim.

```

housing_price_model.py > HousePricePredictor > setup_aws_credentials
26 class HousePricePredictor:
29     def __init__(self):
44         else:
45             self.sagemaker_session = None
46         ...
47     def setup_aws_credentials(self):
48         """AWS credentials setup"""
49         print("🐼 AWS Credentials Setup")
50         print("=" * 40)
51         ...
52         if not AWS_AVAILABLE:
53             print("❌ AWS libraries not installed.")
54             print("Install with: pip install boto3 sagemaker")
55             return False
56         ...
57         # Check if AWS credentials exist
58         try:
59             session = boto3.Session()
60             credentials = session.get_credentials()
61             ...
62             if credentials is None:
63                 print("❌ AWS credentials not found.")
64                 print("\nSetup options:")
65                 print("1. AWS CLI: aws configure")
66                 print("2. Environment variables:")
67                 print("    export AWS_ACCESS_KEY_ID=your_access_key")
68                 print("    export AWS_SECRET_ACCESS_KEY=your_secret_key")
69                 print("    export AWS_DEFAULT_REGION=us-east-1")
70                 print("3. IAM role (if running on EC2/SageMaker)")
71                 return False
72         else:
73             print(f"✅ AWS credentials found")
74             print(f"Region: {session.region_name}")
75

```

Ana kodumda AWS CLI kullanarak SageMaker yapılandırmasını ve konfigürasyonunu ayarlıyorum önce.

Projede XGBoost kullanacağım tablo şeklindeki verilerde iyi sonuç veriyor, AWS üzerinde XGBoost'un hazır bir imajını kullanıyorum.

```

def train_sagemaker_model(self, train_data_path='sagemaker_data/train.csv'):
    """SageMaker'da XGBoost modelini eğit"""
    print("🚀 Training model on SageMaker...")

    if not AWS_AVAILABLE or not self.sagemaker_session:
        print("❌ AWS SageMaker not available")
        return None

    # Get role
    self.role = self.get_sagemaker_role()
    if not self.role:
        print("❌ Cannot proceed without SageMaker role")
        return None

    # S3 bucket check
    if not self.create_s3_bucket():
        print("❌ Cannot proceed without S3 bucket")
        return None

    try:
        # S3'e veri yükle
        s3_input_train = self.sagemaker_session.upload_data(
            path=train_data_path,
            bucket=self.bucket,
            key_prefix='xgboost-train'
        )
        print(f"✅ Data uploaded to: {s3_input_train}")

        # XGBoost estimator
        xgb_estimator = XGBoost(
            entry_point='train.py', # Training script
            role=self.role,
            instance_type='ml.m4.xlarge',
            instance_count=1,
            framework_version='1.5-1',
            py_version='py3',
            hyperparameters={
                'objective': 'reg:squarederror',
                'num_round': 100,
                'max_depth': 6,
                'eta': 0.1,
                'subsample': 0.8,
                'colsample_bytree': 0.8
            }
        )
    
```

Sonra verimi hazırlayıp SageMaker'da eğitmek için olan kodumu çalıştırıyorum.

Aşağıdaki de eğitilirken çalışmasını istediğim kod, bunu da gönderiyorum

```
3 def create_sagemaker_training_script(set):
4     training_script = '''
5
6     import argparse
7     import os
8     import pandas as pd
9     import xgboost as xgb
10    import joblib
11
12    def parse_args():
13        parser = argparse.ArgumentParser()
14        parser.add_argument('--num_round', type=int, default=100)
15        parser.add_argument('--max_depth', type=int, default=6)
16        parser.add_argument('--eta', type=float, default=0.1)
17        parser.add_argument('--subsample', type=float, default=0.8)
18        parser.add_argument('--colsample_bytree', type=float, default=0.8)
19        parser.add_argument('--objective', type=str, default='reg:squarederror')
20
21        parser.add_argument('--model_dir', type=str, default=os.environ.get('SM_MODEL_DIR'))
22        parser.add_argument('--train', type=str, default=os.environ.get('SM_CHANNEL_TRAIN'))
23
24        return parser.parse_args()
25
26    def main():
27        args = parse_args()
28
29        # Veriyi yükle
30        train_data = pd.read_csv(os.path.join(args.train, 'train.csv'), header=None)
31
32        # Target ve features
33        y = train_data.iloc[:, 0]
34        X = train_data.iloc[:, 1:]
35
36        # XGBoost training
37        dtrain = xgb.DMatrix(X, label=y)
38
39        params = {
40            'objective': args.objective,
41            'max_depth': args.max_depth,
42            'eta': args.eta,
43            'subsample': args.subsample,
44            'colsample_bytree': args.colsample_bytree
45        }
46
47        model = xgb.train(params, dtrain, num_boost_round=args.num_round)
48
49        # Model kaydet
50        model.save_model(os.path.join(args.model_dir, 'xgboost-model'))
51
52    if __name__ == '__main__':
53        main()
54    '''
55    return training_script
```

```

Loading and preparing data...
Data shape: (545, 13)
Encoded mainroad: ['no', 'yes']
Encoded guestroom: ['no', 'yes']
Encoded basement: ['no', 'yes']
Encoded hotwaterheating: ['no', 'yes']
Encoded airconditioning: ['no', 'yes']
Encoded prefarea: ['no', 'yes']
Encoded furnishingstatus: ['furnished', 'semi-furnished', 'unfurnished']
Features: ['area', 'bedrooms', 'bathrooms', 'stories', 'mainroad', 'guestroom', 'basement', 'hotwaterheating', 'airconditioning', 'parking', 'prefarea', 'furnishingstatus']
Preparing data for SageMaker...
Data saved for SageMaker training
SageMaker training script created: train.py

Testing SageMaker training...
Training model on SageMaker...
Using SageMaker role: arn:aws:iam::257394496046:role/SageMakerExecutionRole-HousePrice
S3 bucket creation failed: An error occurred (BucketAlreadyExists) when calling the CreateBucket operation: The requested bucket name is not available. The bucket namespace is shared by all users of the system. Please select a different name and try again.
Trying with bucket name: house-price-prediction-bucket-1947
S3 bucket 'house-price-prediction-bucket-1947' created
Data uploaded to: s3://house-price-prediction-bucket-1947/xgboost-train/train.csv
Starting SageMaker training job...
INFO:sagemaker:Creating training-job with name: sagemaker-xgboost-2025-05-25-10-59-21-715
2025-05-25 10:59:29 Starting - Starting the training job...
2025-05-25 10:59:52 Starting - Preparing the instances for training...
2025-05-25 11:00:25 Downloading - Downloading input data...
2025-05-25 11:00:56 Downloading - Downloading the training image.....
2025-05-25 11:02:17 Training - Training image download completed. Training in progress.../miniconda3/lib/python3.8/site-packages/xgboost/compat.py:36: FutureWarning: pandas.Int64Index is deprecated and will be removed from pandas in a future version. Use pandas.Index with the appropriate dtype instead.
from pandas import MultiIndex, Int64Index

```

S3 Bucket kullanarak önce train datayı bir yere kaydediyor, ardından eğitim tamamlanıyor.

```

from pandas import MultiIndex, Int64Index
/miniconda3/lib/python3.8/site-packages/xgboost/data.py:262: FutureWarning: pandas.Int64Index is deprecated and will be removed from pandas in a future version. Use pandas.Index with the appropriate dtype instead.
elif isinstance(data.columns, (pd.Int64Index, pd.RangeIndex)):

2025-05-25 11:02:46 Uploading - Uploading generated training model
2025-05-25 11:02:46 Completed - Training job completed
Training seconds: 140
Billable seconds: 140
SageMaker training completed!
SageMaker training started successfully!
(base) Mustafa-3:mlproje mustafagirgin$

```

Eğitim tamamlandıktan sonra oluşan modeli başka bir S3 bucket'a kaydediyor dağıtım için. Ben direkt api ile SageMaker üzerinden tahminimi yapabiliyorum.


```
}'  
{ "predicted_price": 9968810.0, "currency": "INR", "formatted_price": "₹9,968,810.00", "features_used": { "area": 8000, "bedrooms": 4, "bathrooms": 2, "stories": 3, "mainroad": "yes", "guestroom": "yes", "basement": "no", "hotwaterheating": "no", "airconditioning": "yes", "parking": 2, "prefarea": "yes", "furnishingstatus": "furnished" }, "model_type": "sagemaker_endpoint", "model_version": "2.0.0" }
```

```
~/desktop/nlpproje > █
```

15:18:29